

DATA 505: High-Performance Distributed Datastores

Term: Fall 2026 | Instructor: Dr. Jeffrey Dean (jdean@systems.cloud.edu)

Course Description & Overview:

Architecture of consensus protocols, log-structured storage engines, columnar query engines, and distributed transaction semantics.

COURSE ASSIGNMENTS & WEIGHT DISTRIBUTION

Assignment Description	Weight	Due Module	Deliverable Format
Distributed Raft Consensus Engine	30%	Module 05	Go/Rust Cluster Repo & Tests
LSM-Tree Key-Value Storage Engine	30%	Module 09	Benchmarked C++ Storage Engine
Distributed Query Planner Capstone	30%	Module 12	Distributed Engine Showcase
Code Review & Architecture Critique	10%	Continuous	Bi-Weekly Pull Request Audits

WEEKLY TERM SCHEDULE & READING REQUIREMENTS

Timeline	Technical Topic Focus	Assigned Readings & Docs
Week 01	Module 01: Storage Engine Fundamentals & B-Trees	Kleppmann (Ch. 3); Graefe (2011)
Week 02	Module 02: Log-Structured Merge (LSM) Trees	O'Neil et al. (1996); RocksDB Architecture Specs
Week 03	Module 03: Distributed Consensus & Raft Protocol	Ongaro & Ousterhout (2014)
Week 04	Module 04: Paxos vs Multi-Paxos Architectures	Lamport (1998); Chandra et al. (2007)
Week 05	Module 05: Columnar Formats & Apache Parquet	Apache Parquet Documentation
Week 06	Module 06: Transactions & Two-Phase Commit	Bernstein et al. (Ch. 7)
Week 07	Module 07: Spanner: TrueTime & Consistency	Corbett et al. (2012)
Week 08	Module 08: Messaging with Apache Kafka	Kreps et al. (2011); Kafka Core Architecture
Week 09	Module 09: Vector Search & Distributed Indexes	Malkov & Yashunin (2018)
Week 10	Module 10: Stream Processing with Apache Flink	Akidau et al. (2015)
Week 11	Module 11: CRDTs and Conflict-Free Replication	Shapiro et al. (2011)
Week 12	Module 12: Final Query Engine Presentations	Systems Benchmarks Report